

Project: AI Disaster Damage Assessment & Response System

When a disaster such as an **earthquake, flood, landslide, or wildfire** occurs in Nepal, responders often need to understand which areas and buildings are affected before sending limited resources.

Our system would take **before-and-after satellite or aerial images** of a disaster-affected area and use computer vision to automatically identify buildings and classify their condition as **undamaged, possibly damaged, or severely damaged**.

The results would appear on an interactive **map**, where each affected building is highlighted. The system could then calculate the approximate number of damaged buildings, identify the most severely affected zones, and generate a simple **priority map for emergency responders**.

Instead of replacing human disaster-response teams, the AI would act as a **first-pass assessment tool**. Human officials could review and correct the AI results before they are used for actual decisions. This follows the same human-in-the-loop approach used by the UN's system.

For Nepal, make it more useful

Add a Nepal-specific **Response Priority Score**:

Damage severity + population density + critical infrastructure + accessibility = response priority

For example, if an earthquake damages 500 buildings, the system should not simply show all 500. It could highlight:

- **Red:** severe damage + high population
- **Orange:** moderate damage
- **Yellow:** suspected damage
- **Blue:** hospitals, schools, bridges and other critical infrastructure nearby

The dashboard could then tell responders:

Ward 8: 127 potentially damaged buildings
34 severely damaged
Estimated affected population: 2,400
3 critical facilities nearby
Priority: Very High

This turns the project from simply **"AI detects damaged buildings"** into **"AI helps authorities decide where to respond first."**

Simple technical architecture

Satellite imagery → Image preprocessing → Building detection → Damage classification → GIS/map → Priority scoring → Disaster-response dashboard

For a hackathon MVP, you do not need to recreate Google's entire system. Their real solution itself combines building detection, damage models, satellite imagery, and expert validation.

A realistic MVP would focus on **one disaster type**, such as earthquakes or floods, and demonstrate the complete pipeline on a selected Nepal location.

Week 1 — Research, Planning & Data Preparation

During the first week, our main focus will be to **understand the existing research, define the Nepal-specific problem, and prepare the data and technical foundation** for the project.

1. **Study existing research**
 - Analyze the Google Research and UN approach to AI-based disaster damage assessment.
 - Identify what methods can realistically be adapted for Nepal.
2. **Define the MVP**
 - Initially focus on **one disaster type**, preferably earthquakes.
 - Define the output: damaged buildings, damage severity, affected areas, and response priority.
3. **Collect and prepare data**
 - Find suitable pre- and post-disaster satellite/aerial imagery.
 - Identify publicly available Nepal disaster datasets and building/location data.
 - Organize and preprocess the selected images.
4. **Design the AI pipeline**
 - Plan the workflow: **Satellite Image → Building Detection → Damage Classification → Map → Priority Score**.
 - Decide which computer-vision models and GIS technologies will be used.
5. **Build the initial prototype structure**
 - Set up the project repository and development environment.
 - Create the basic map/dashboard interface.
 - Prepare the model-training and data-processing pipeline.

Week 1 goal: By the end of the first week, we should have a clearly defined MVP, relevant datasets identified and prepared, the technical architecture finalized, and the initial dashboard/model pipeline ready for development.